

Unit 4

Question bank

Short Answer Questions

1. Name and briefly describe two common classification metrics used to evaluate a model.
2. Describe the main steps involved in the c-means clustering algorithm.
3. What is cluster validity and why is it important in clustering analysis?
4. Mention two indices used to measure cluster validity and briefly describe one.
5. How can you choose the optimal number of clusters in FCM?
6. Describe how fuzzy logic can be combined with neural networks for multi-feature pattern recognition
7. Explain classification by equivalence relation in the context of fuzzy set theory.
8. How does equivalence relation help in grouping data objects in classification?
9. Define intra-cluster similarity and inter-cluster dissimilarity.
10. What is the key difference between fuzzy c-means clustering and k-means clustering?
11. What is the significance of choosing the number of clusters (k) in cluster analysis?
12. What is the effect of overlapping partitions in feature space on classification accuracy?

Long answer questions

1. Explain the concept of cluster validity and discuss different internal and external validity indices used for evaluating clustering results. Provide examples of how these indices help in selecting the optimal number of clusters.
2. Discuss the advantages and limitations of Fuzzy C-Means clustering. In what types of datasets is FCM particularly useful, and when might it fail or perform poorly?
3. Derive the objective function of the Fuzzy C-Means algorithm and explain the iterative update steps for membership values and cluster centers. Why does FCM converge, and what are the stopping criteria?

4. What are the common issues faced by the K-Means algorithm? Explain the problems of initialization, local minima, and cluster shape assumptions. Suggest possible solutions or improvements to address these challenges.
5. Define cluster analysis and discuss its importance in data mining and machine learning. Explain the main types of clustering techniques and provide examples of applications for each.
6. Describe the role of single sample identification in fuzzy system and machine learning. How can a single data instance be correctly identified or classified, and what techniques are used to improve accuracy?
7. Describe the basic components and working of a fuzzy classification system. How are membership functions defined and used to assign degrees of belonging to different classes? Provide examples of common membership functions.
8. Discuss the role of fuzzy logic in pattern recognition. How does fuzzy logic help in improving the robustness and flexibility of pattern classifiers? Illustrate with examples from image recognition or speech recognition systems.